

Assignment: Correlation and Regression Analysis on Network Performance Data

Scenario:

You are analyzing performance data from 10 different wireless network nodes deployed in a smart campus environment. For each node, the following data were recorded:

- **Bandwidth (Mbps)**
- **Latency (ms)**
- **Packet Loss (%)**
- **Energy Consumption (Watts)**

You are tasked with understanding how bandwidth and latency affect performance and energy usage.

Dataset:

Node ID	Bandwidth (Mbps)	Latency (ms)	Packet Loss (%)	Energy Consumption (W)
1	12	50	1.2	2.8
2	18	45	1.0	3.2
3	10	60	1.6	2.5
4	25	30	0.7	4.0
5	22	35	0.9	3.7
6	30	25	0.5	4.5
7	28	28	0.6	4.3
8	20	40	1.1	3.4
9	15	48	1.3	3.0
10	26	27	0.6	4.1

 Tasks:

1. Correlation Analysis (6 Marks)

a) Calculate the **Pearson correlation coefficients** for the following variable pairs:

- Bandwidth vs Latency

b) Comment on the **relationship direction** (positive/negative) and **strength** (weak/moderate/strong).

2. Scatter Plots (6 Marks)

a) Create **scatter plots** for:

- Latency vs Packet Loss

b) Add a **trend line** and describe any pattern (linear/nonlinear, increasing/decreasing).

3. Simple Linear Regression (8 Marks)

a) Fit regression models to:

- Predict **Energy Consumption** based on **Bandwidth**

b) For each:

- Write the **regression equation**
- Report the **R-squared** value
- Interpret the **slope** and discuss what it implies for network performance

✓ **Bandwidth (Mbps)**

- **Meaning:** The maximum amount of data that can be transmitted over a network connection in one second.
- **Unit: Megabits per second (Mbps)**
- **Example:** A 25 Mbps connection can transfer up to 25 million bits per second.
- **Interpretation: Higher bandwidth = faster data transfer.**

✓ Latency (ms)

- **Meaning:** The time it takes for a data packet to travel from the source to the destination.
- **Unit: Milliseconds (ms)**
- **Example:** If latency is 50 ms, it takes 0.05 seconds for data to travel one way.
- **Interpretation: Lower latency = faster response times (better for real-time applications like video calls or gaming).**

✓ Packet Loss (%)

- **Meaning:** The percentage of data packets that are sent but never reach the destination.
- **Unit: Percent (%)**
- **Example:** If 1% of packets are lost, that means 1 out of every 100 data packets gets dropped.
- **Interpretation: Lower packet loss = more reliable connection.**

✓ Energy Consumption (Watts)

- **Meaning:** The amount of electrical power consumed by the device or system.
- **Unit: Watts (W)**
- **Example:** A device consuming 3.5 W is using 3.5 joules of energy per second.
- **Interpretation: Lower energy consumption = more energy-efficient system.**